

Data basics

INTRODUCTION TO DATA



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Data is everywhere

Education level

Temperature

Social media post

Name

Age

Sex

Home address

Favorite grocery store

Coffee cups

GDP rate

Height

Weight

House dimensions

Eye color

Number of shirts

Number of schools

Distance Earth-Moon

What is data?

- Derived from **datum**: *given, fact*
- Valuable resource in this digital era¹
- Two main formats of data (simplified):
 - Structured
 - Unstructured



¹ The Economist, May 6th 2017: The world most valuable resource is no longer oil, but data

Unstructured vs. structured data

Student record - unstructured format

There are five students named James, Elizabeth, Thomas, David, and Sarah who scored the maximum grade of 10. Their respective genders and ages are male 16, female 14, male 17, male 15 and female 11.

Unstructured vs. structured data

Student record - unstructured format

There are five students named James, Elizabeth, Thomas, David, and Sarah who scored the maximum grade of 10. Their respective genders and ages are male 16, female 14, male 17, male 15 and female 11.

Student record - structured format

Name	Age	Gender	Grade
James	16	Male	10
Elizabeth	14	Female	10
Thomas	17	Male	10
David	15	Male	10
Sarah	11	Female	10

Structured data

- Organized, easy to manage
- Tabular format
- Predefined structure
- Text and numbers

Examples:

- Customer information
- Financial transactions

Student record - structured format

Name	Age	Gender	Grade
James	16	Male	10
Elizabeth	14	Female	10
Thomas	17	Male	10
David	15	Male	10
Sarah	11	Female	10

Unstructured data

- Unorganized, difficult to manage
- No specific format
- No predefined structure
- Text, images, audio, video

Examples:

- Reports and email messages
- Surveillance videos

Student record - unstructured format

There are five students named James, Elizabeth, Thomas, David, and Sarah who scored the maximum grade of 10. Their respective genders and ages are male 16, female 14, male 17, male 15 and female 11.

Quantitative vs. qualitative data

Quantitative data

- Also called numerical data
- Count, measure, represent with numbers

Examples:

- Height of a person or object (e.g. 184 cm)
- Room temperature (e.g. 25 °C)

Qualitative data

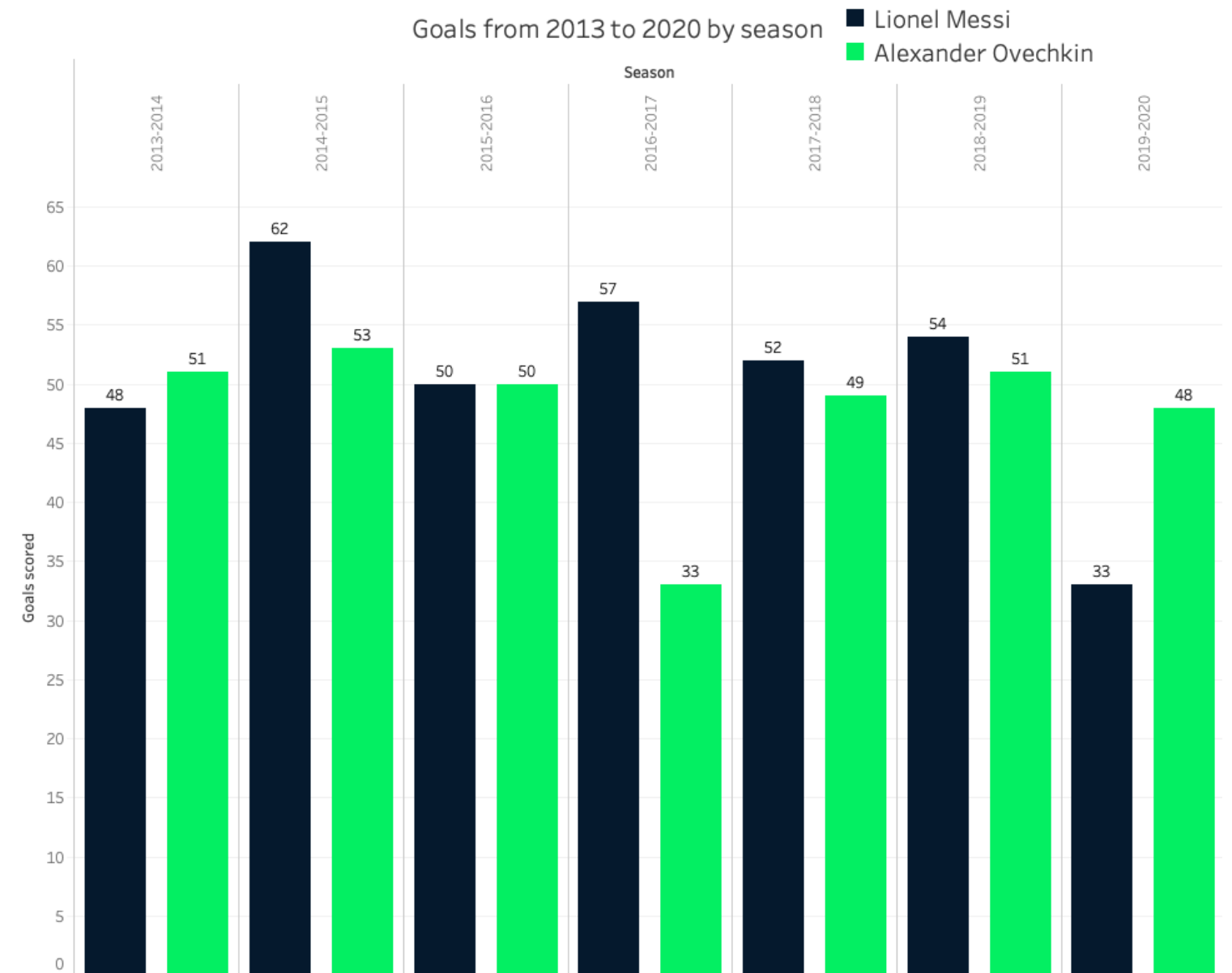
- Also called categorical data
- Group into categories

Examples:

- Language (English, French, Arabic)
- Favorite holiday destination (Spain, France, Greece)

Data context

- Who is a great player?
 - Lionel Messi
 - Alexander Ovechkin



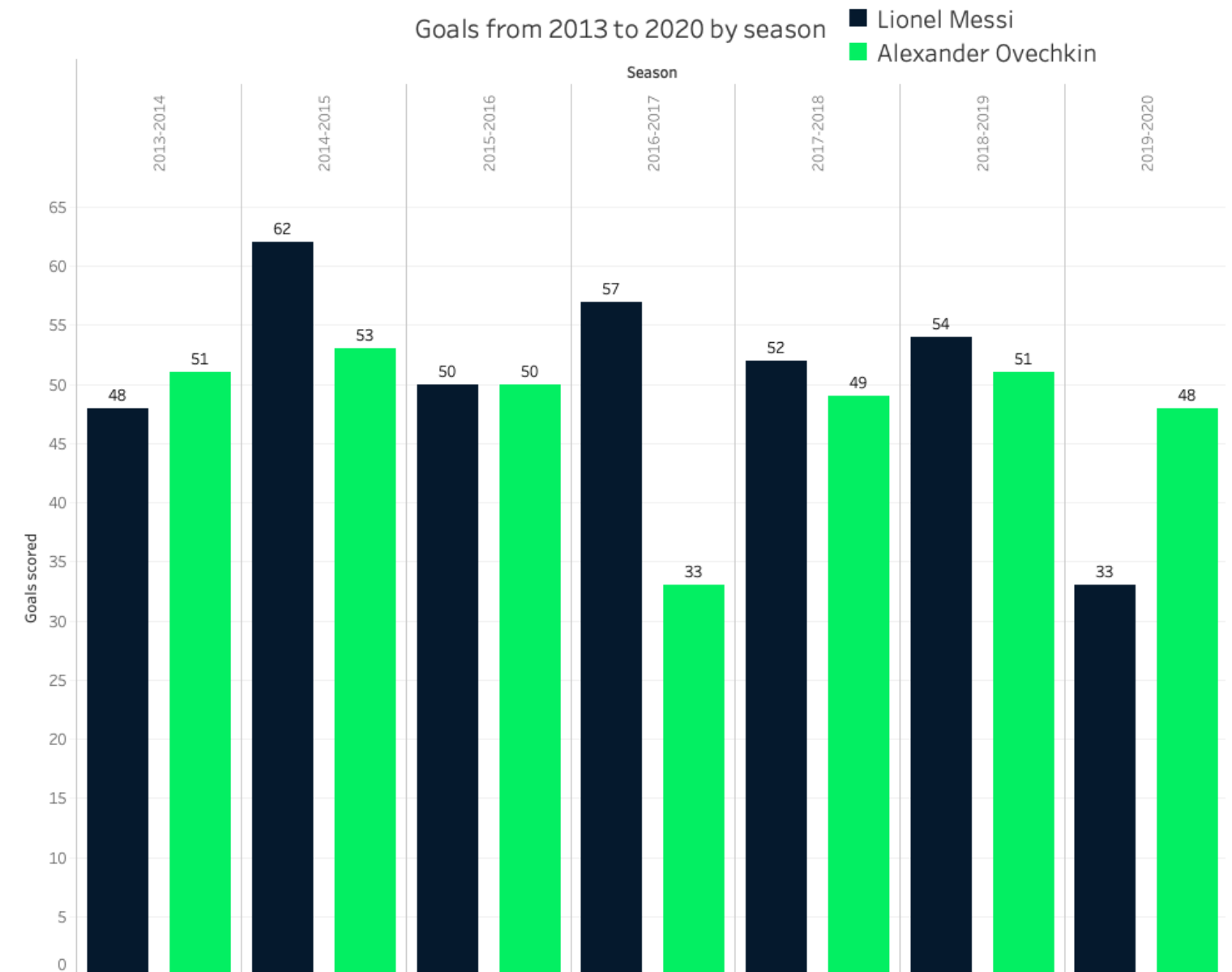
Data context

Information that provides meaning to data

- Characteristics of the data
 - Time frame
 - Location and source

These characteristics of the data are called the **metadata**

- Industry, sports, research, or business knowledge



Let's practice!

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The value of data

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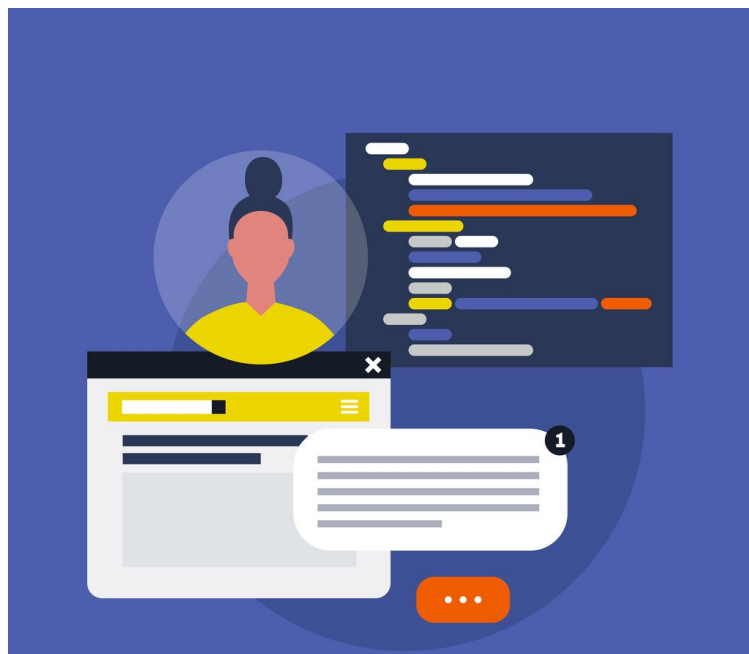
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Individuals and organizations use data

Data for individuals

- Personal finances
- Time management
- Health tracking
- ...



Data for organizations

- Government & non-profit
- Healthcare
- Supply Chain
- ...



Data in organizations

Goal: support business objectives

- Profitability
- Social good
- Research
- Customer satisfaction & employee happiness

How: by improving decision making

- Measure return on investment (ROI)
- Optimize processes and find new opportunities



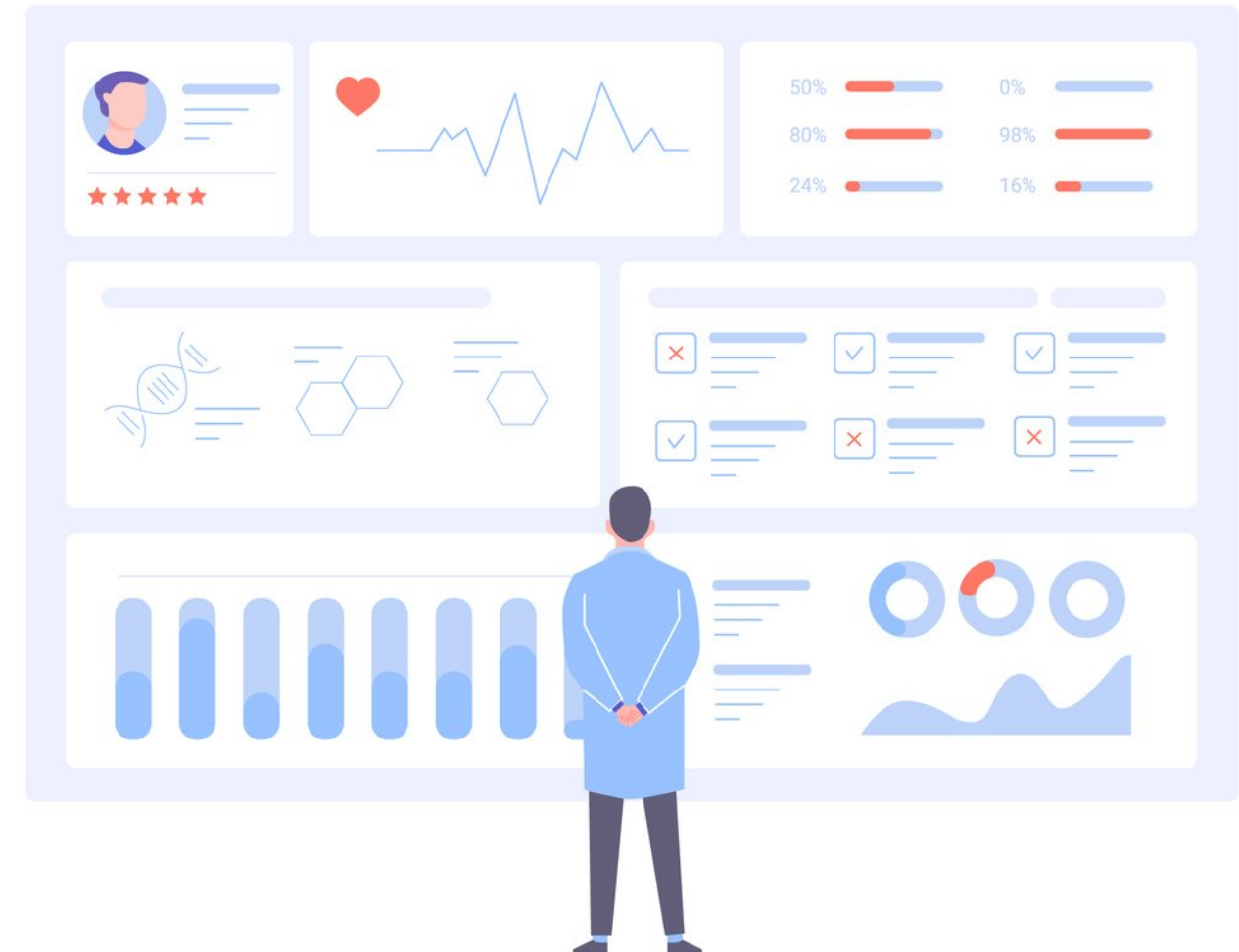
Data in healthcare

Example: wearable devices

- Monitor personal data

Goal:

- Detect and prevent health problems
- Turning patient care into precision medicine
- Advancing healthcare research worldwide



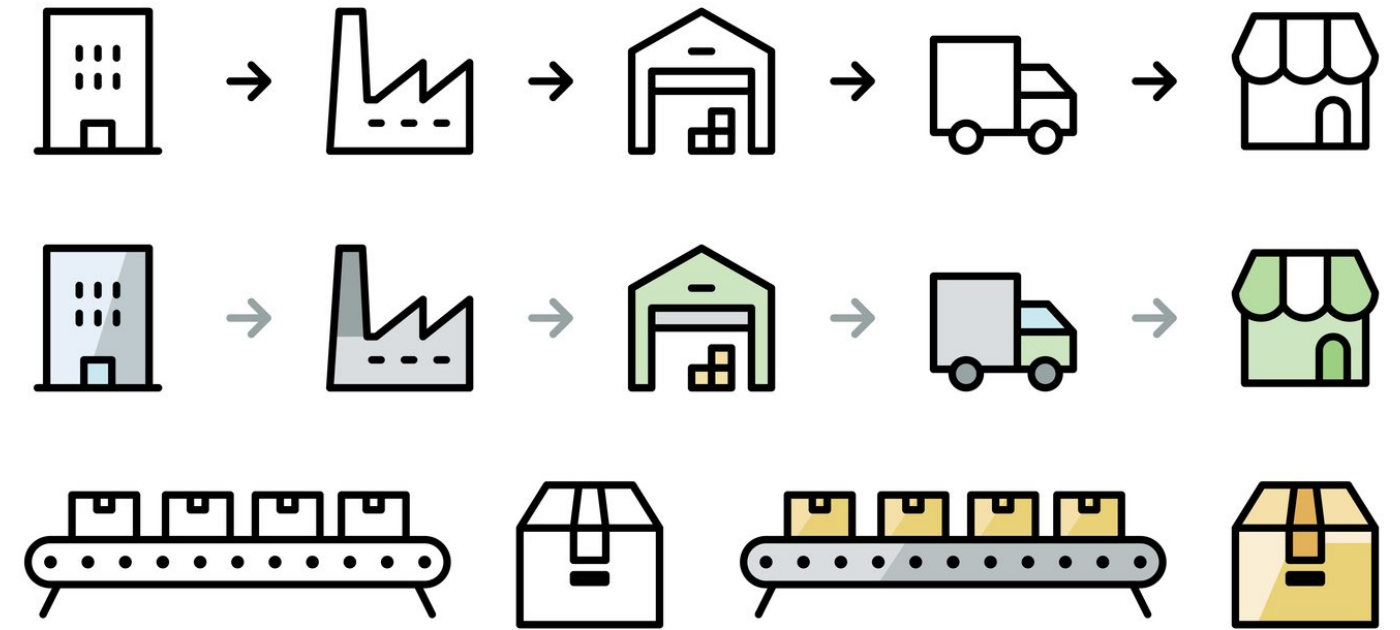
Data in supply chain

Example: monitoring various metrics

- Average inventory
- Inventory turnover ratio

Goal:

- Make sense of the massive amount of generated data
- Demand forecasting



The sequence of processes involved in the production and distribution of a product¹.

¹ Oxford Languages

Data in education

Example: DataCamp courses

- User feedback
- Incorrectly submitted answers
- Learner drop-off

Goal:

- Improve course design
- Identify struggle areas
- Personalize learning experience

Data is a competitive advantage

Why is that?

- Unique to an organization
- Unavailable for competitors
- Generate insights and actions based on that data



Let's practice!

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The curious case of data growth

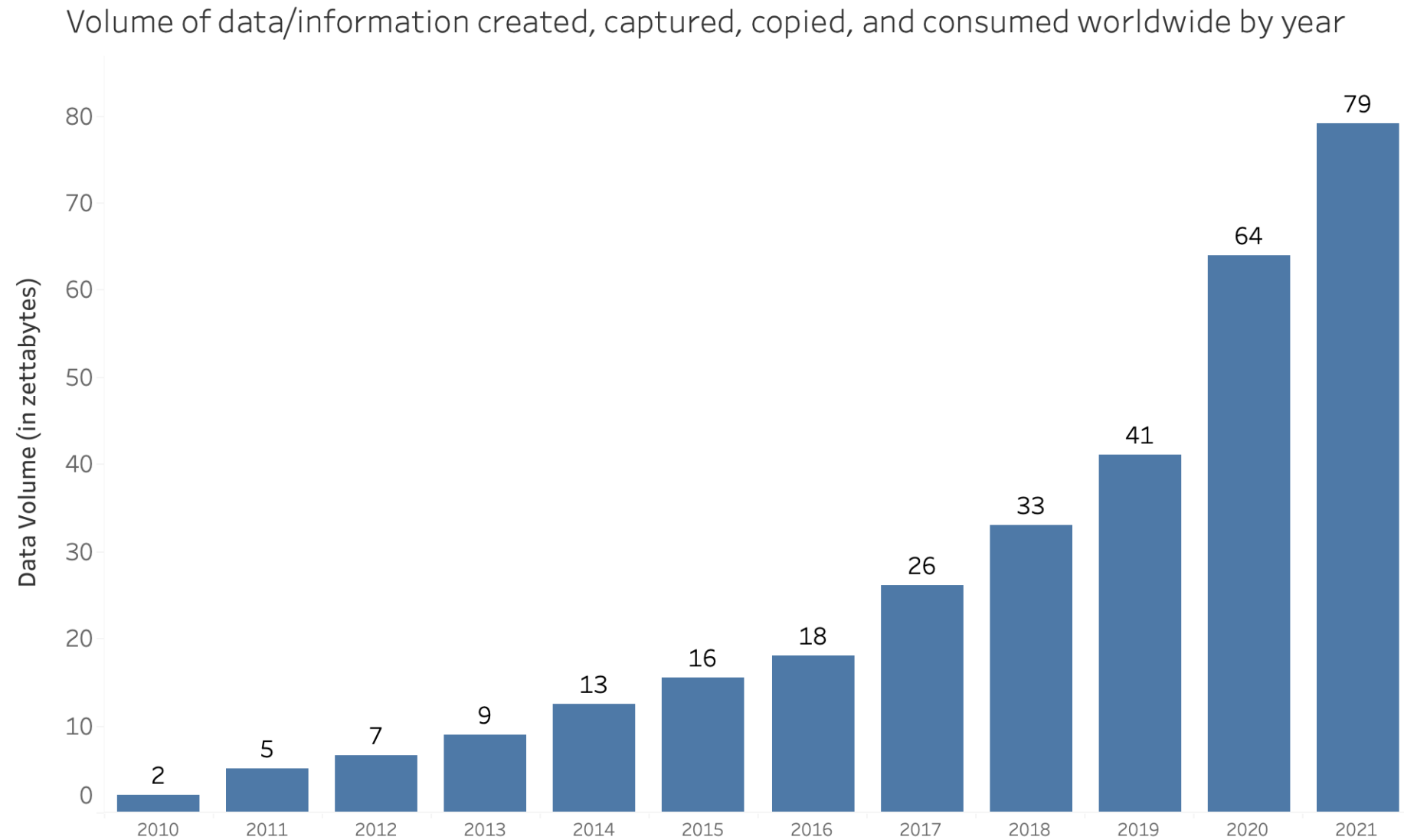
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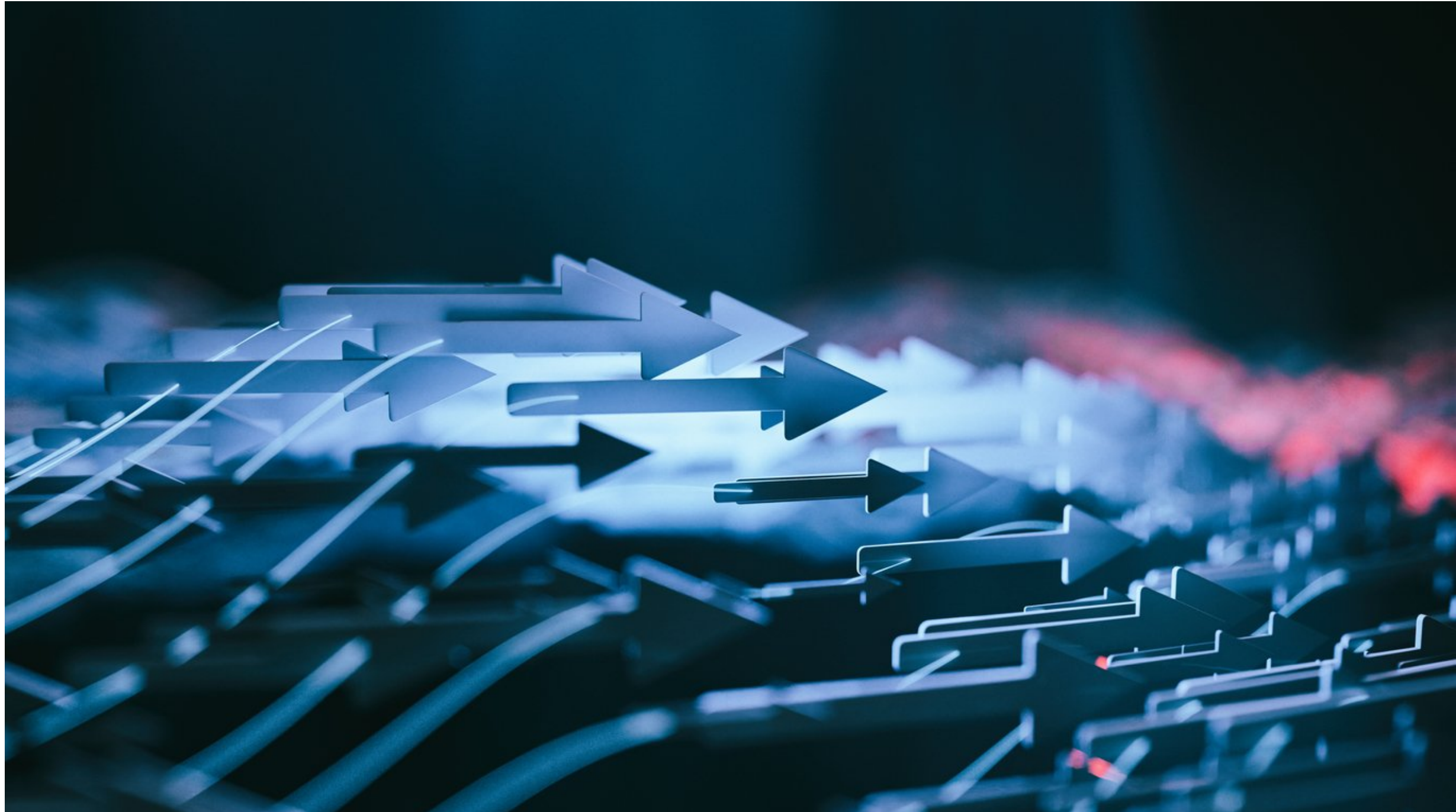
The volume of data has grown exponentially



1 zettabyte = a one followed by 21 zero's in bytes = 1 billion terrabyte

¹ Source: Statista

Data storage is changing



Data storage is changing

Historical data storage

- Genetic information in DNA
- Cave and wall paintings
- Scrolls and books of papyrus/parchment



19th and 20th century

- Punch cards
- Magnetic tape, floppy disks



20th and 21st century

- More data on smaller media
- CDs and hard/solid state drives (local)
- Data centers (cloud)



Data - where does it come from?

Ice cream shop #1 in New York



- Sells vanilla, chocolate, and strawberry
- Has a rough idea of sale transactions

Ice cream shop #2 in New York



- Sells 20+ ice cream flavors
- Also sells coffees and milkshakes
- Tracks all sales

Capturing data

Data captured

- Sales per product type and ice cream flavor
- Stock per product type and flavor
- Weather data

Optimizations

- Avoid popular flavors being out of stock
- Replace poor selling flavors with new ones
- Predict sale spikes due to high temperature
- Optimize prices

Ice cream shop #2 in NY, USA



Which ice cream shop would fare better?

Ice cream shop #1 in New York



- Uses gut feeling to make decisions
- Randomly switches ice cream flavors

Ice cream shop #2 in New York



- Uses data to make decisions
- Searching for the best flavors

Companies are more complex than ice cream shops

3D Manufacturing companies

- Beam heat
- Layer thickness
- Structural stability



Financial institutions

- Mortgage applications
- Fraud detection



Let's practice!

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